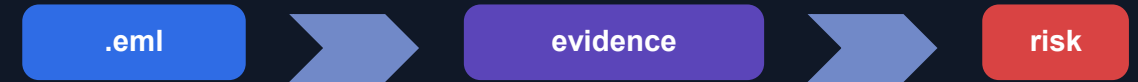


PHISHSHIELD

Local and Explainable Phishing Email Triage

with Deterministic Analysis and Experimental AI
Assistance



safe · local · explainable

**A practical security workflow with a
responsible path toward AI assistance.**

José Nieto Alaber

Máster en Desarrollo con IA · Big School
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A Reported Email Needs A Safe First Look

Reported .eml

Users forward suspicious messages for review. The original message may contain URLs, attachments, and sensitive context.



Unsafe first inspection

Opening the message in a normal mail client can expose the analyst to the very content they need to investigate.



Evidence before escalation

A help-desk or SOC analyst needs a repeatable first assessment, not an opaque verdict.

The first triage step should be safe, local, and explainable.

Local-First Triage For Analysts And Help-Desk Teams

PHISHSHIELD

Local email triage workbench

Upload an .eml message and inspect the local backend risk posture, grouped indicators, and current evidence summary.

Analysis input

Load a local email and send it to the current backend triage flow.

EMAIL FILE

Choose or drag a .eml message to send to `/api/analyze-email`.

No file selected

Privacy: This deployment processes email through the configured backend. Use a trusted local or internal deployment for confidential email. Do not upload confidential or production email to a public or untrusted host.

Choose .eml file

Analyze email

Execution path

Browser upload to `/api/analyze-email` proxied to the local FastAPI service through the frontend proxy.

Current scope

The MVP focuses on local .eml triage, risk scoring, grouped findings, and a readable evidence summary.

No analysis loaded

Upload an .eml file to render the current backend findings, risk score, and grouped indicators.

LOCAL-FIRST

EXPLAINABLE

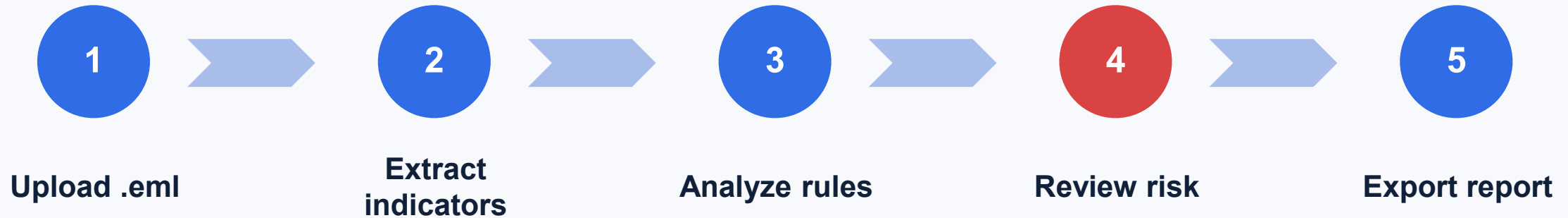
DETERMINISTIC

EXPORTABLE

A safe and repeatable first look before escalation.

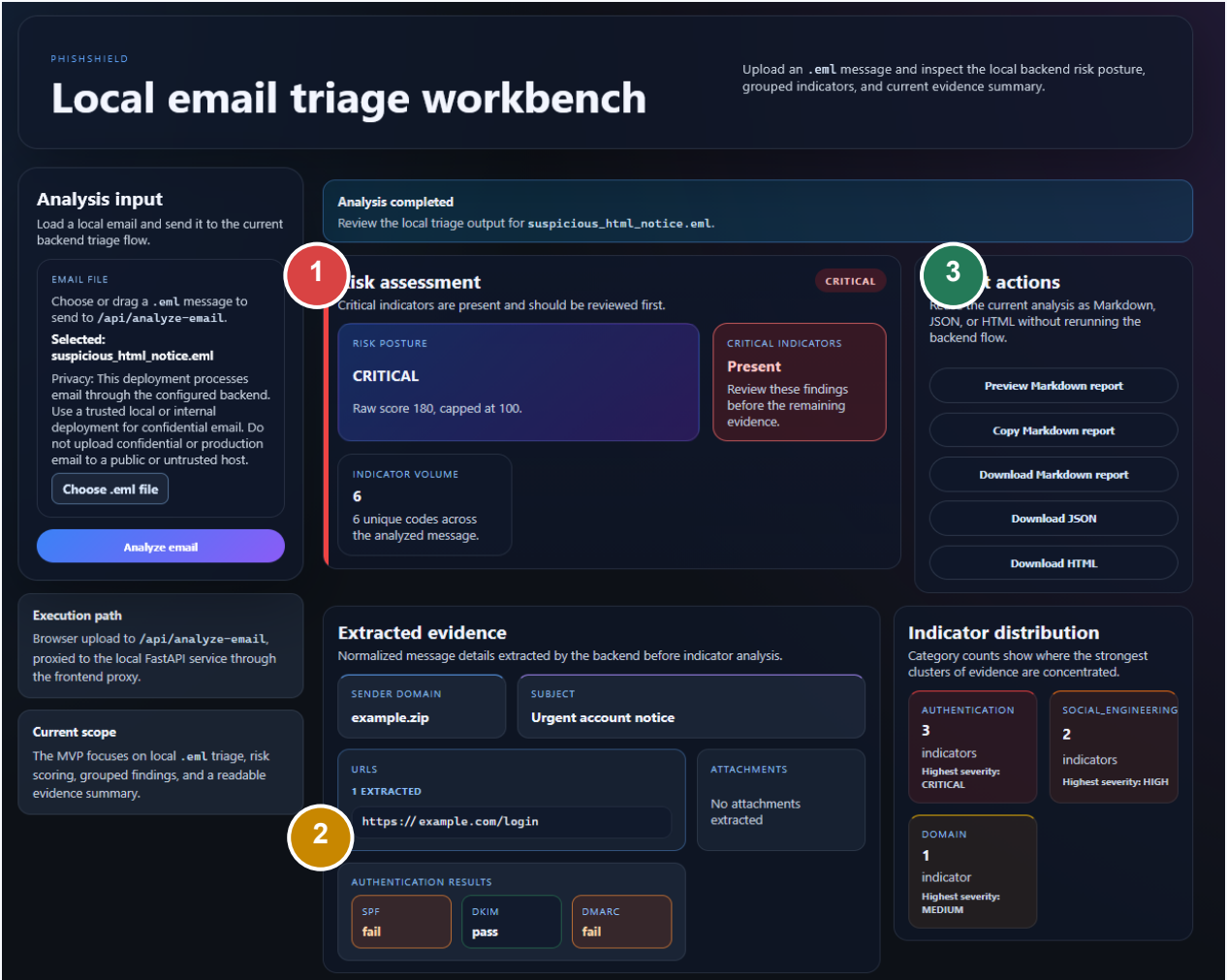
PhishShield processes the message locally, explains the evidence behind each finding, and produces a deterministic risk assessment.

From Reported Message To Evidence



Technical evidence · grouped findings · deterministic risk · reusable reports

How To Read An Analysis Result



1. Deterministic Risk

CRITICAL is calculated from the combined deterministic findings.

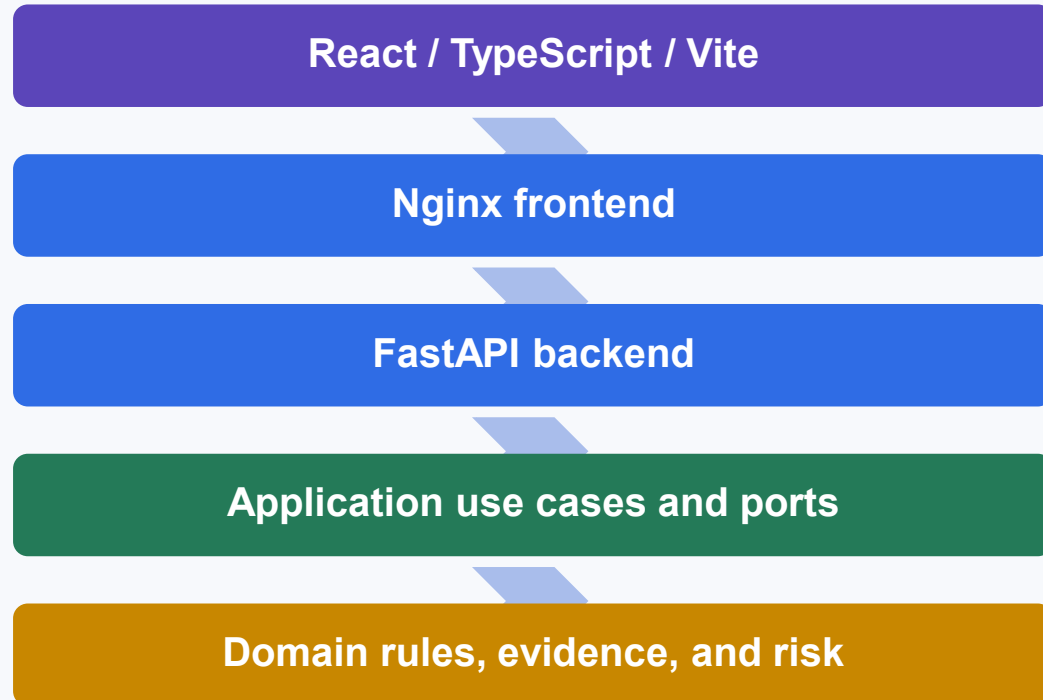
2. Header Authentication

SPF, DKIM, and DMARC are extracted from email headers. They are technical evidence, not AI predictions.

3. Report Actions

Export the current evidence as Markdown, JSON, or HTML.

A Modular Monolith With Clear Boundaries



Backend

Python · FastAPI · Pydantic

Frontend

React · TypeScript · Vite

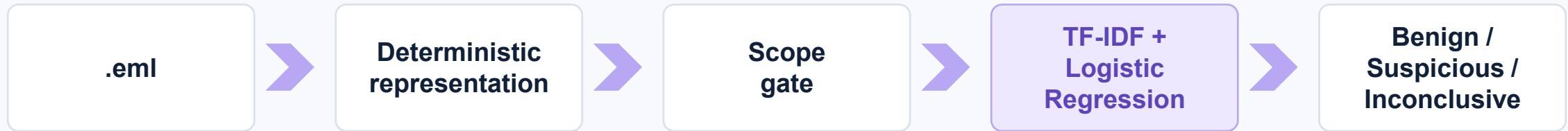
Distribution

Docker · Compose · GHCR

Quality

pytest · Vitest · GitHub Actions · pre-commit

A Local Advisory Capability, Not An Automated Verdict



The deterministic path remains authoritative for findings, evidence, risk, and CI decisions.

Evidence Quality Sets The Promotion Boundary

What is not sufficient yet

- Authorized data
- Representative families
- Independent validation
- Temporal evaluation
- Reliable false-positive and false-negative analysis

Safety decision

- Local processing
- Explicit abstention
- Deterministic-first results
- No automated production verdict
- Clear promotion criteria for future work

Limiting the model is a safety decision, not an absence of work.

A Public, Reproducible Foundation For Further Validation



Current result

A functional local security tool with reproducible execution.

Current boundary

No representative authorized corpus for AI promotion.

Next step

Independent validation with privacy-preserving data.

PhishShield demonstrates a practical security tool and a responsible path toward AI-assisted triage.